

2021

Time : 3 hours

Full Marks : 40

*Candidates are required to give their answers
in their own words as far as practicable.*

The figures in the margin indicate full marks.

Answer all sections as directed..

Section-A

(Compulsory)

1. Pick up the correct alternative for each of
the following questions : $10 \times 1 = 10$

- (a) In tossing three coin at a time, the
probability of getting at most one head
is :

(i) $\frac{3}{8}$

(ii) $\frac{7}{8}$

(iii) $\frac{1}{8}$

☒ (iv) $\frac{1}{2}$

(b) A card is drawn at random from a well shuffled pack of 52 cards. The probability that the card is a king or a queen is :

☒ (i) $\frac{2}{13}$

(ii) $\frac{3}{13}$

(iii) $\frac{4}{13}$

(iv) $\frac{8}{13}$

(c) The probability of a certainty is equal to :

(i) 0

☒ (ii) 1

(iii) ∞

(iv) None of these

(d) The probability can take value :

(i) $-\infty$ to ∞

(ii) 1 to ∞

(iii) -1 to 1

☒ (iv) 0 to 1

(e) Mean is a measure of :

(i) Dispersion

☒ (ii) Location

(iii) Correlation

(iv) None of these

(f) If a constant value 10 is subtracted from each observation of a set, then the mean of set is :

(i) Increased by 10

☒ (ii) Decreased by 10

(iii) Not affected

(iv) Zero

(g) The correct relationship between AM, GM and HM is :

(i) $AM = GM = HM$

(ii) $GM \geq AM \geq HM$

(iii) $HM \geq GM \geq AM$

☒ (iv) $AM \geq GM \geq HM$

(h) Extreme value have no effect on :

(i) Average

☒ (ii) Median

- (iii) Geometric Mean
- (iv) Harmonic Mean
- (i) The term regression was introduced by :
- (i) R A Fisher
- ☒ (ii) Sir Francis Galton
- (iii) Karl Pearson
- (iv) None of these
- (j) The value of correlation coefficient varies from :
- (i) 0 to 1
- ☒ (ii) -1 to 1
- (iii) -1 to 0
- (iv) 0 to ∞

Section-B

Answer any **three** questions :

3×5=15

2. State addition and multiplication theorem of probability.
3. Discuss statistical and mathematical definition of probability.
4. Define arithmetic mean with properties.
5. How will you calculate median in case of grouped data.
6. Define correlation coefficient.
7. Discuss about binomial distribution.

Section-C

Answer any **two** questions of the following :

2×7.5=15

8. There are two bags I and II. Bag I contains 3 white and 4 black balls and bag II contains 5

white and 6 black balls. One ball is drawn at random from one of the bags and is found to be white. Find the probability that it was drawn from bag I.

9. Calculate the mean from the following table :

Class interval : 30-39 40-49 50-59 60-69 70-79 80-89

Frequency : 5 22 63 74 30 6

10. Explain Baye's theorem with examples.

11. Define regression coefficient and how it differs from correlation.

_____ x _____